

Simulation of bit stuffing on a binary string

as data link layer data framing method on sender machine side

- framing flag is 01111110
- using bit_stuffing method prevents to creation a substring in string that look like flag's binary pattern thats why on receiver sides de-framing frames is correctly available
- in string converts every 11111 substrings to 111110

```
def bit_stuffing(string):  
    result = ""  
    counter = 0  
  
    for i in string:  
        result += i  
  
        if i == "1":  
            counter += 1  
  
            if counter == 5:  
                result += "0"  
                counter = 0  
        else:  
            counter = 0  
  
    return result
```

Simulation of bit stuffing

on a binary string as data link layer data framing method on receiver machine side

- in string converts every 111110 substrings to 11111
- bit destuffing on receiver side is required ot recovery true and main stuffed data

```
def bit_destuffing(string):  
    result = ""  
    counter = 0  
    i = 0  
  
    while i < len(string):  
        bit = string[i]
```

```

result += bit

if bit == "1":
    counter += 1

    if counter == 5:
        if i + 1 < len(string) and string[i + 1] == "0":
            i += 1 # skips bit `0` value to write on result
            counter = 0
    else:
        counter = 0

i += 1

return result

```

Getting input from user and some checks with print logs

- check that input maximum has 100 bits
- check that input is a binary string value

```

# enter a binary string as input would you simulate base on it
data = "0110111111111111111111110010"

```

```

if len(data) > 100:
    print("Error: Input size must be less than 101 bits.")
    print_result_bool = False

elif not all(bit in "01" for bit in data):
    print("Error: Input must contain only 0 and 1.")
    print_result_bool = False

else:
    print_result_bool = True

```

Results of simulation methods based on user input

```

if print_result_bool:
    stuffed_data = bit_stuffing(data)
    destuffed_data = bit_destuffing(stuffed_data)

```

```
print("Original data:      ", data)
print("Bit stuffed data:  ", stuffed_data)
print("Bit destuffed data:", destuffed_data)

if data == destuffed_data:
    print("Data recovery was successful.")
else:
    print("Data recovery failed.")

Original data:      011011111111111111110010
Bit stuffed data:   011011111011111011111010010
Bit destuffed data: 011011111111111111110010
Data recovery was successful.
```